
accu-positioner

OPERATION AND MAINTENANCE MANUAL

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Contents

WARRANTY.....	4
Customer Assistance	5
Change Record	6
1. Introduction	7
2. Product Specifications	8
3. Installation.....	9
3.1. Electrical	9
3.2. Mounting.....	9
4. Operation	12
4.1. GUI Operation	12
4.2. Serial Operation	12
4.2.1. ROS Helios Serial Operation	12
4.2.2. ROS Visca Serial Operation	14
4.3. Networking Multiple Positioners	20
5. Maintenance	22
5.1. Remove and Replace the Yoke Bracket and Yoke Ring	22
5.2. Remove the Mounting Bracket and Stop Ring	22
6. Troubleshooting.....	24
6.1. Systematic TroubleShooting Procedure.....	24
6.2. Unit Slips or drops the load.....	25
6.3. Unit Does Not Respond to Commands	25
6.4. Unit Rotates in Unexpectedly Short Range of Travel.....	25
7. Parts List	26

Figures

Figure 1. AccuPositioner	7
Figure 2. Minimum Measured Torque vs. Speed	8
Figure 3. Sample Wiring	9
Figure 4. Installation Outline with Yoke and Mounting Block	10
Figure 5. Installation Outline without Yoke and Mounting Block	11
Figure 6. Multiple Units Connected to Multiple Ports	21
Figure 7. Stop Collar Install	22
Figure 8. Mounting Bracket Install	23
Figure 9. Exploded View	27

Tables

Table 1. Mechanical Specifications.....	8
Table 2. Electrical Specifications	8
Table 3. Serial Parameters.....	12
Table 4. Helios Commands and Inquiries	12
Table 5. Visca Commands and Inquiries	15
Table 6. Serial Response Messages	20
Table 7. Parts List.....	26

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Change Record

<u>REV</u>	<u>DESCRIPTION</u>	<u>DATE</u>
A	SEE EC-03199	08JUN20
B	SEE EC-03239	27AUG20
C	SEE EC-03265	20OCT20
D	SEE EC-03424	28OCT21

1. Introduction

The AccuPositioner (Figure 1) is a high-accuracy, high torque, light-weight, compact positioner for underwater use. Options include oil-filled or air-filled. The AccuPositioner is powered by 20-28 VDC to provide 10 ft lb of torque, and can turn up to 24 deg/s, 360 deg continuous.



Figure 1. AccuPositioner

2. Product Specifications

Table 1 shows the mechanical specifications for the AccuPositioner. Table 2 shows the electrical specifications.

Table 1. Mechanical Specifications

	Air-Filled	Oil-Filled
Weight in air	6.6 lb (3.0 kg)	7.7 lb (3.5 kg)
Weight in water	3.1 lb (1.4 kg)	4.2 lb (1.9 kg)
Depth rating	230 ft (70 m)	20 k ft (6.0 km)
Operating temp	-10 to 50°C	
Storage temp	-20 to 60°C	
Material	Housing: 6061-T6 Shaft: 17-4 PH Yoke: 5052-H32 Mtg block: 6061-T6 or Delrin	
Range	Continuous	
Maximum speed	24 deg/s	
Torque	See Figure 2	

Table 2. Electrical Specifications

Voltage	20 - 28 VDC
Current	2 A Max
Communication	RS 485 Half-Duplex, 300 - 115200 Baud
Accuracy	+/- 0.1 deg

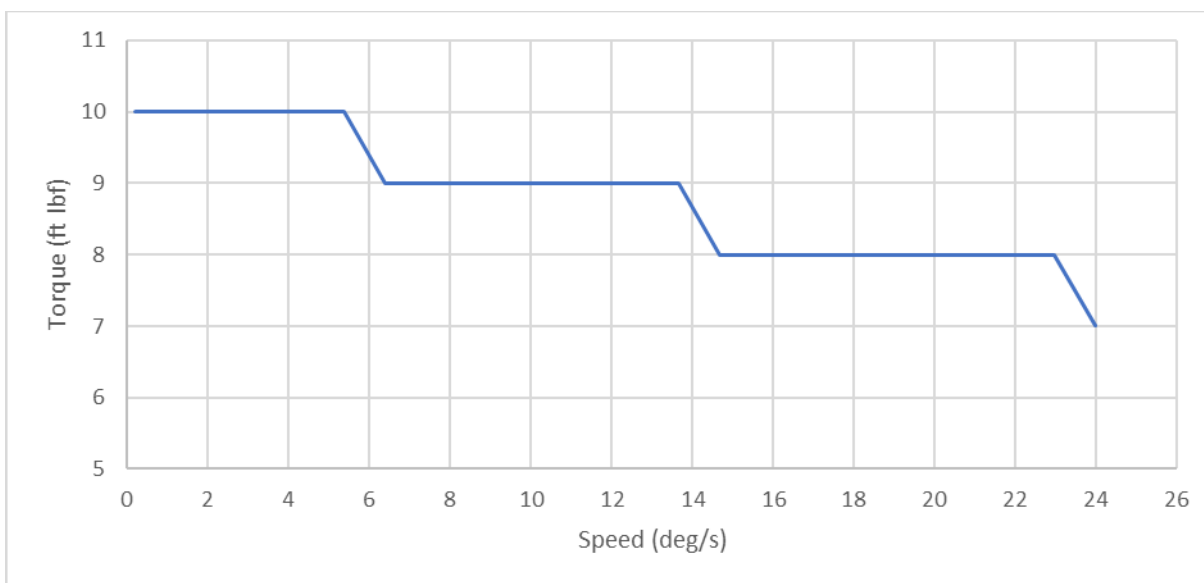


Figure 2. Minimum Measured Torque vs. Speed

3. Installation

3.1. Electrical

The AccuPositioner pinout will depend on customer selections; Figure 3 shows the default wiring.

3.2. Mounting

Figure 4 shows the outline of the AccuPositioner with the yoke and mounting base sold by Remote Ocean Systems. Figure 5 shows the outline of the AccuPositioner without the yoke and mounting base. The AccuPositioner may be mounted by the pan or tilt axis shaft.

 WARNING

The AccuPositioner should be operated only through a circuit with ground fault interrupt protection.

24VDC RETURN	—————	1
24VDC SUPPLY	—————	2
RS485 +	—————	3
RS485 -	—————	4

Figure 3. Sample Wiring

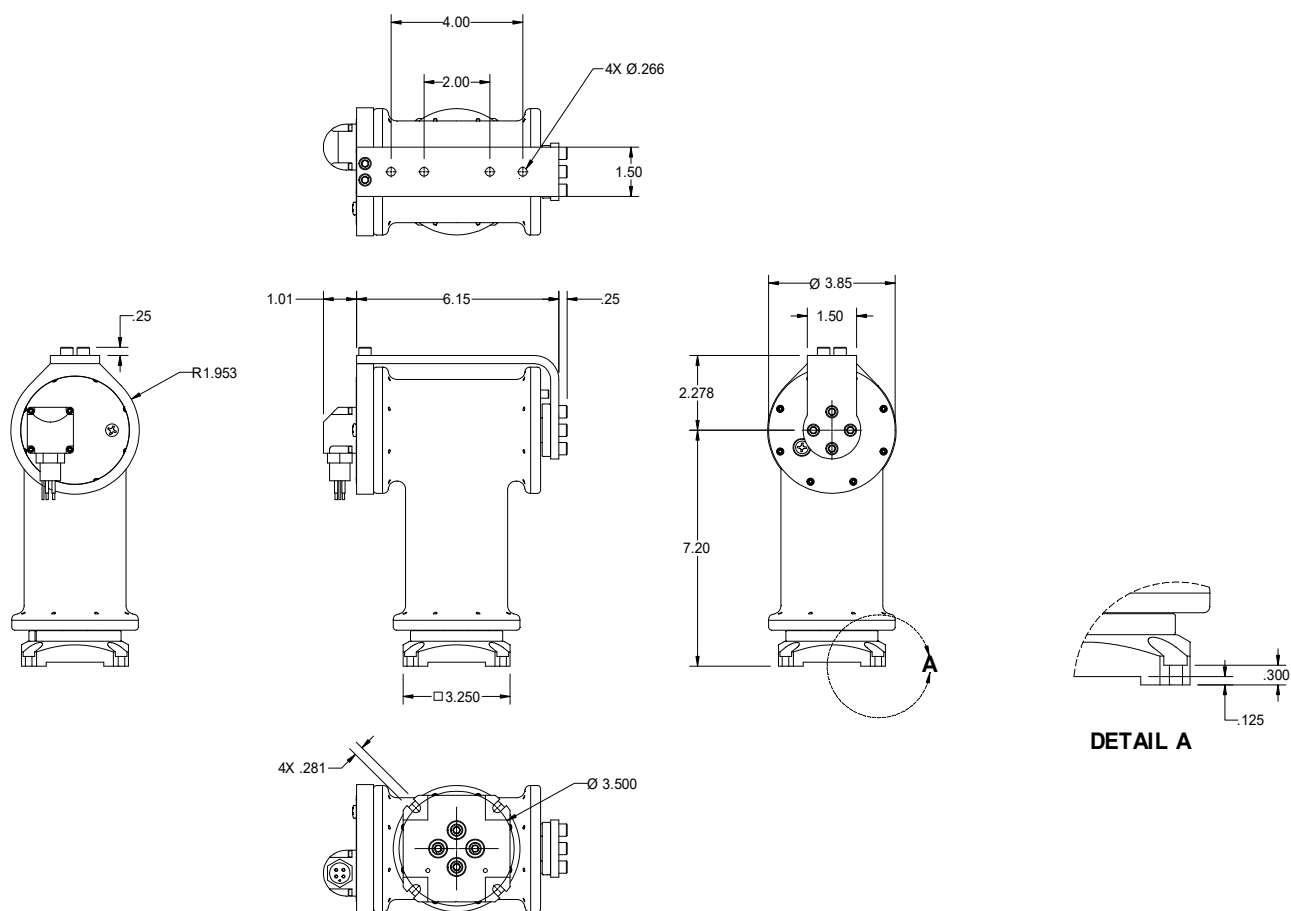


Figure 4. Installation Outline with Yoke and Mounting Block

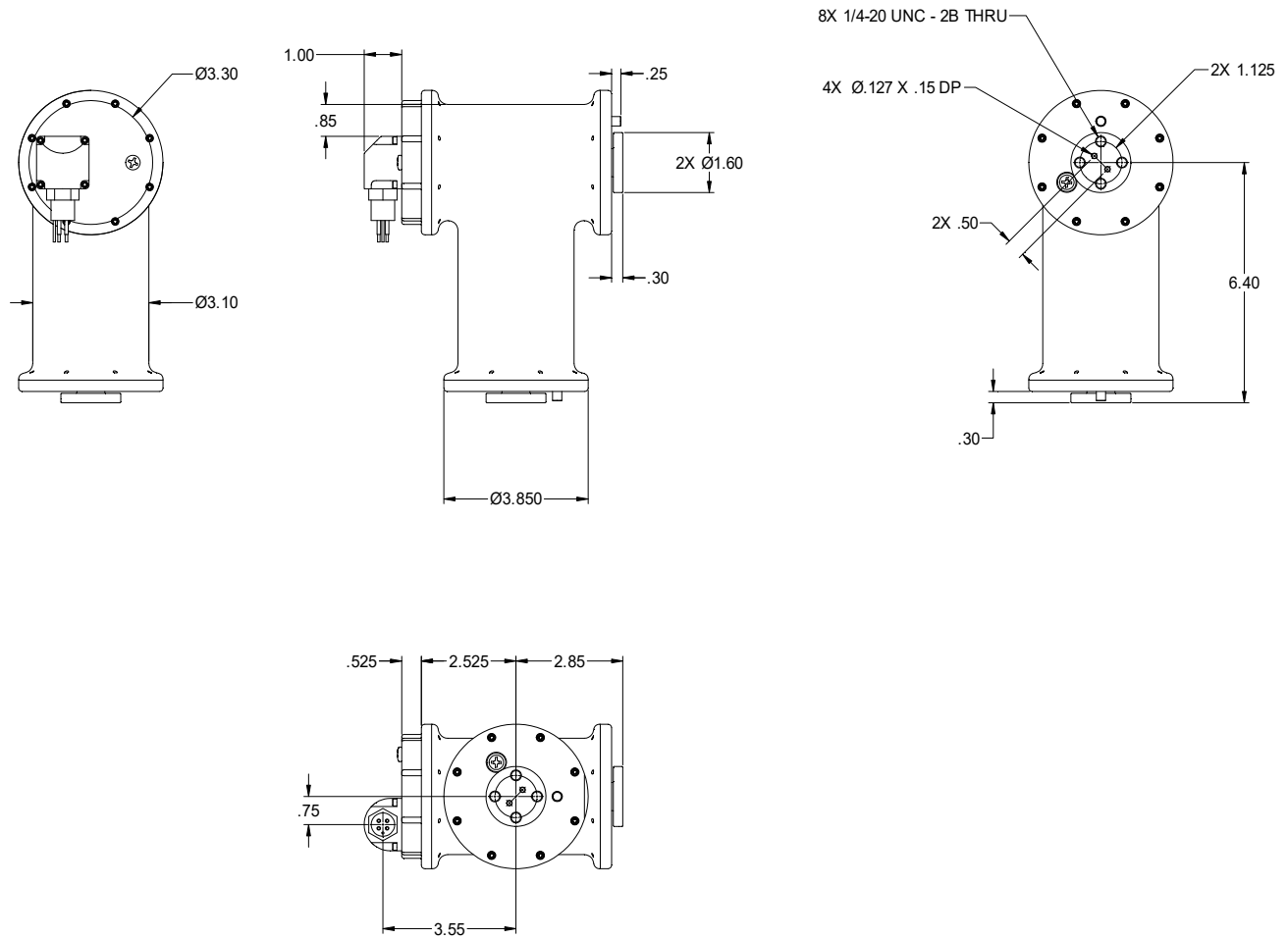


Figure 5. Installation Outline without Yoke and Mounting Block

4. Operation

4.1. GUI Operation

The AccuPositioner can be operate using the ROS provided GUI, Eos. Please contact ROS for access to the GUI and documentation.

4.2. Serial Operation

The AccuPositioner can be operated by by sending RS-485 serial commands directly from user-written software or firmware. There are two modes, Visca and Helios. Each mode has a command to switch to the other. Table 3 shows the AccuPositioner serial communications parameters applicable to both modes. Helios mode is described in 4.2.1, Visca mode is described in 4.2.2.ROS Helios Serial Operation.

Table 3. Serial Parameters

Baud Rate	9600 (default)
Data bits	8
Start bits	1
Stop bits	1
Parity	None
Flow control	None

4.2.1. ROS Helios Serial Operation

The AccuPositioner in Helios mode functions as two separate addressable nodes. The node addresses shipped from the factory are 'A' (pan) and 'B' (tilt). Commands are addressed to the pan and tilt nodes separately. Node addresses can be changed by the user to facilitate networking. For instance, if two positioners are on the same RS485 bus, the first unit may be addressed as A and B, and the second as C and D. All Helios commands are ASCII format. Valid addresses are from 'A' (0x65) to '' (0x96). Table 4 shows the available Helios commands. All commands shown use address 'A' as an example. The address of the actual node should be substituted when sending the commands.

Table 4. Helios Commands and Inquiries

Command	Format	Example	Notes
Drive Closed Loop	{node}p{val}	Ap150	{val} = RoundUp({angle (1 - 359.5)} / (360 / ({factoryccwlimit} - {factoryccwlimit})) + {factoryccwlimit} + 0.5)
Get Acceleration level	{node}?003	A?003	Returns {node}{value}. Acceleration level = ({value} + 1)*2 deg/s ² . {value} is 000, 001, 002, 003, 004
Get Braking Value	{node}?006	A?006	Returns {node}{value}. 0 ≤ {value} ≤ 128
Get Calibrated Position	{node}g	Ag	Deprecated--Returns {node}{pos} (see ClosedLoopPosition)

Table 4. Helios Commands and Inquiries

Command	Format	Example	Notes
Get Serial Delay	{node}?002	A?002	Returns {node}{value}. {value} 000 - 999, multiples of .25 ms, e.g., 010 = 2.5 ms. Serial delay is between receive character and echo, or between receive character and response. No delay between sent characters in a string.
Get Character Echo Status	{node}?001	A?001	Returns <node>e<value>. Value = 001 (enabled) or 000 (disabled). Echoes unit address, good and bad characters. Does not echo valid address that is not the unit address.
Get Factory Settings	{node}?000	A?000	Returns {id:0},{fact_ccw_limit:2-4},{fact_cw_limit:6-8},{user_ccw_limit:10-12},{user_cw_limit:14-16},{pcb_dash_no:18},{pos_feedback:20},{pcb_ser_no:22-25},{baud:27},{type:29},{firmware_rev:31-32}
Get Moving Flag	{node}?007	A?007	Returns {node}{value}, {value} is 001 (moving), or 000 (not moving)
Get Position	{node}f	Af	Returns {node}{pos}, {pos} = RoundUp({angle} * (1 - 359.5) / (360 / ({factoryccwlimit} - {factoryccwlimit})) + {factoryccwlimit} + 0.5)
Get Slip Stall Flag	{node}?005	A?005	Returns {node}{value}, {value} is 001 (slip/stall), or 000 (no slip/stall)
Get Step Count	{node}q	Aq	Returns {node}{+/-}{value}, value is 5-digit step count from last counter reset
Get Velocity Limit	{node}?004	A?004	Returns {node}{value} = 3-digits, 2x limit in deg/s, e.g., 003 = 1.5 deg/s
Reset Step Counter	{node}z000	Az000	Resets step counter to zero
Rotate CCW	{node}<{val}	A<003	{val} = 3-digits, 2x desired speed in deg/s, e.g., 003 = 1.5 deg/s
Rotate CW	{node}>{val}	A>003	{val} = 3-digits, 2x desired speed in deg/s, e.g., 003 = 1.5 deg/s
Rotate a Number of Steps	{node}y{dir}{spd}{steps}	A y 0 12 00035	{dir} = 0(ccw) or 1 (cw), {spd} = 00 - 48 = 2x speed in deg/s, {steps} = number of steps 00001 - 65536
Rotate CW with Acceleration Ramp	{node}+{val}	A+003	{val} = 3-digits, 2x desired speed in deg/s, e.g., 003 = 1.5 deg/s
Rotate CCW with Acceleration Ramp	{node}-{val}	A-003	{val} = 3-digits, 2x desired speed in deg/s, e.g., 003 = 1.5 deg/s
Set Acceleration Level	{node}a{val}	Aa001	Acceleration level = ({value} + 1)*2 deg/s ² . {value} is 000, 001, 002, 003, 004

Table 4. Helios Commands and Inquiries

Command	Format	Example	Notes
Set Counterclockwise Rotation Limit	{node}d{val}	Ad150	{val} = RoundUp({angle} (1 - 359.5)) / (360 / ({factorycwlmit} - {factoryccwlmit})) + {factoryccwlmit} + 0.5)
Set Clockwise Rotation Limit	{node}u{val}	Au150	{val} = RoundUp({angle} (1 - 359.5)) / (360 / ({factorycwlmit} - {factoryccwlmit})) + {factoryccwlmit} + 0.5)
Set Serial Delay	{node}b{val}	Ab004	{val} = 000 - 999, multiples of .25 ms, e.g., 010 = 2.5 ms
Set Character Echo Status	{node}e{val}	Ae001	{val} = 001 (enable), or 000 (disable). Echoes unit address, good and bad characters. Does not echo valid address that is not the unit address.
Set Node ID	{node}i{val}	Ai001	{val} = 001 - 032
Set Velocity Limit	{node}m{val}	Am003	{val} = 3-digits, 2x desired limit in deg/s, e.g., 003 = 1.5 deg/s
Rotate One Step	{node}z{val}	Az001	{val} = 001 CW, 002 CCW, 000 clear counter Each step = 0.0102272727272727 deg
Stop with Braking Power	{node}s{val}	As000	{val} = 000 (max) to 128 (min)
Stop with Deceleration Ramp	{node}t	At	

4.2.2. ROS Visca Serial Operation

The AccuPositioner can be operated in Visca Mode by sending RS-485 serial commands directly from user-written software or firmware. Table 5 lists commands and inquiries that may be sent to the AccuPositioner, along with their expected responses and behavior.

Table 6 lists standard response messages.

NOTE

All Visca commands, inquiries and responses are terminated with hex byte ff (255).

Table 5. Visca Commands and Inquiries

Command	Packet	Note
Cancel Command	8x 2y ff	Stops the positioner from driving open loop or closed loop without deceleration. Returns Command Cancelled x = address y = socket (default 1)
Drive Open Loop	8x 01 06 01 vv ww yy zz ff	x = address vv: Pan speed 0x01 (low speed) to 0x18 (high speed) or 0x00 ww: Tilt speed 0x01 (low speed) to 0x00 (high speed) or 0x00 yy = direction: 01 = left, 02 = right, 03 = none zz = direction: 01 = up, 02 = down, 03 = none Speed range is 0.2 (0x01) to 24 (0x18) deg/s. The positioner responds Command Acknowledged after receiving and executing a valid command. It will ramp up to the set speed and run until receiving Cancel, Drive Open Loop, or Drive Absolute Position. To stop the unit with ramping, send drive open loop command with 0 speed.

Table 5. Visca Commands and Inquiries

Command	Packet	Note
Drive Absolute Position	8x 01 06 02 vv ww 0y 0y 0y 0y 0z 0z 0z 0z ff	x = address vv: Pan speed 0x01 (low speed) to 0x18 (high speed) ww: Tilt speed 0x01 (low speed) to 0x18 (high speed) yyyy: Pan Position zzzz: Tilt Position Speed range is 0.2 (0x01) to 24 (0x18) deg/s. Position range is 0 (0 deg) to $2^{16} - 1$ (360 deg - epsilon) bits. The unit responds Command Acknowledged after receiving and beginning a valid command. The unit will servo to the set position to the best of its ability without going faster than the set speed or accelerating faster than the maximum acceleration. If a set point is commanded outside of the user limits, the unit will servo to the user limit. The positioner will continue to servo until it receives Cancel, Drive Open Loop, or Drive Absolute Position. To stop the unit with ramping, send drive open loop command with 0 speed. If position limits are cleared, the unit will take the shortest route to the set point. If position limits are set, the unit will take the route that does not pass through zero.
Drive Home	8x 01 06 04 ff	Drive Absolute Position to the User Zero at 5 deg /s.
Set Position Limits	8x 01 06 07 00 0w 0y 0y 0y 0y 0z 0z 0z 0z ff	w: 1 = up right, 0 = down left yyyy: Hex pan limit position zzzz: Hex tilt limit position Set the user limits on position. Position range is 0 (0 deg) to $2^{16} - 1$ (360– 360/2^{16} deg) bits. Returns Command Completed once the limits have been successfully written to non-volatile memory. Returns Not Executable if the unit is already outside the requested limits, the unit is moving, or there is an error writing the limits to non-volatile memory.
Clear Position Limits	8x 01 06 07 01 0w 07 0f 0f 0f 0f 0f 0f 0f ff	x = address w: ignored Clear the user limits on position and allow the unit to rotate continuously. Returns Command Completed once non-volatile memory has been successfully written. Return not executable if there is an error writing to non-volatile memory.

Table 5. Visca Commands and Inquiries

Command	Packet	Note
Get Max Speed	8x 09 06 11 ff	x = address Returns y0 50 ww zz ff ww = Pan max speed (deg/s) zz = Tilt max speed (deg/s) y = address + 8
Get Position	8x 09 06 12 ff	x = address Returns z0 50 0v 0v 0v 0v 0w 0w 0w 0w ff vvvv = Pan position www = Tilt position z = address + 8 Gets the current position relative to the user-set zero. Position range is 0 (0 deg) to $2^{16}-1$ (360 deg - epsilon) bits. Convert to degrees by the formula $x = vvvv * 360 / 2^{16}$
Get Position Limits	8x 09 06 13 0w ff	x = address w: 1 = up right, 0 = down left Returns z0 50 0u 0u 0u 0u 0v 0v 0v 0v FF uuuu = Hex pan limit vvvv = Hex tilt limit z = address + 8 Get the user limits on position. Position range is 0 (0 deg) to $2^{16} - 1$ (360 - $360/2^{16}$ deg) bits. Convert to degrees by the formula $x = uuuu * 360 / 2^{16}$
Get Max Acceleration	8x 09 06 14 ff	x = address Returns z0 50 yy zz yy = Pan max acceleration (1 - 125 deg/s²) zz = Tilt max acceleration (1 - 125 deg/s²) Gets the user-set acceleration limit in degrees/second².

Table 5. Visca Commands and Inquiries

Command	Packet	Note
Get Serial Delay	8x 09 06 52 ff	x = Address Returns z0 50 0y 0y yy = Serial delay Get the current delay between sent bytes in increments of 250 micro seconds.
Set Max Acceleration	8x 01 06 51 0y 0y 0z 0z ff	x = address yy = 8-bit integer Pan max acceleration (0 - 125 deg/s²) zz = 8-bit integer Tilt max acceleration (0 - 125 deg/s²) Sets the user-set acceleration limit in degrees/second². Returns Command Completed once the limits have been successfully written to non-volatile memory. Returns Not Executable if there is an error writing to nonvolatile memory. Returns Syntax Error if greater than 125 is requested max acceleration is requested. Set max acceleration to 0 to allow unlimited acceleration.
Set Serial Delay	8x 01 06 52 0y 0y ff	x = address yy = serial delay / 250 microseconds (0 - 125) Set a delay before the AccuPositioner responds in increments of 250 micro seconds. Returns Command Completed once the limits have been successfully written to non-volatile memory. Returns not executable if there is an error wring to nonvolatile memory.

Table 5. Visca Commands and Inquiries

Command	Packet	Note
Set Zero	8x 01 06 55 0v 0w 0y 0y 0y 0y 0z 0z 0z 0z ff	x = address v = 0 (subtract value from current pan zero), 1 add value to current pan zero) w = 0 (subtract value from current tilt zero), 1 add value to current tilt zero) yyyy = value to add to pan zero zzzz = value to add to tilt zero Set the user-zero to a position offset from the current zero. Offset range is 0 (0 deg) to $2^{16}-1$ (360 - $360/2^{16}$ deg). Returns Command Completed once the zero has been successfully written to non-volatile memory. Returns Not Executable if the requested zero is outside of user limits, the requested zero is greater than 65386 or less than 182, the unit is moving, or there is an error writing to nonvolatile memory.

Table 6. Serial Response Messages

Response	Packet	Note
Command Acknowledged	Z0 4y ff	z = address + 8 y = socket (default 1) Returned when a command that takes time to execute is accepted and begun.
Command Completed	z0 5y ff	z = address + 8 y = socket (default 1) Returned when a command that can be executed instantaneously is completed
Syntax Error	z0 60 02 ff	z = address + 8 Returned when the command format is incorrect or when a command with illegal command parameters is accepted.
Command Not Executable	z0 6y 41 ff	z = address + 8 y = socket (default 1) Returned when a command cannot be executed because it would put the unit out of limits, because the unit's current state makes the command illegal, or because an error occurred.

4.3. Networking Multiple Positioners

The AccuPositioner can be set to answer to one of 7 addresses, 0x81 to 0x87. 0x88 is reserved in the Visca protocol for broadcast commands. The factory default address is 0x81. Positioners on the same network must be set to different addresses, or they will step on each other when trying to respond to commands. Units not connected to the same network may share an address. Consequently, each network may contain up to 7 units. There is no limit to the number of networks. A sample network diagram is shown in Figure 6.

Note

The unit is designed to operate as a device in an RS485 network. Ensure the proper terminating resistor value is used for multiple device networks.

The unit address may be changed using the serial command in Table 5, or via the user settings window in the Eos GUI.

Note

When assigning an address, only one unit may be connected to the network. Otherwise, all connected units will be assigned the new address.

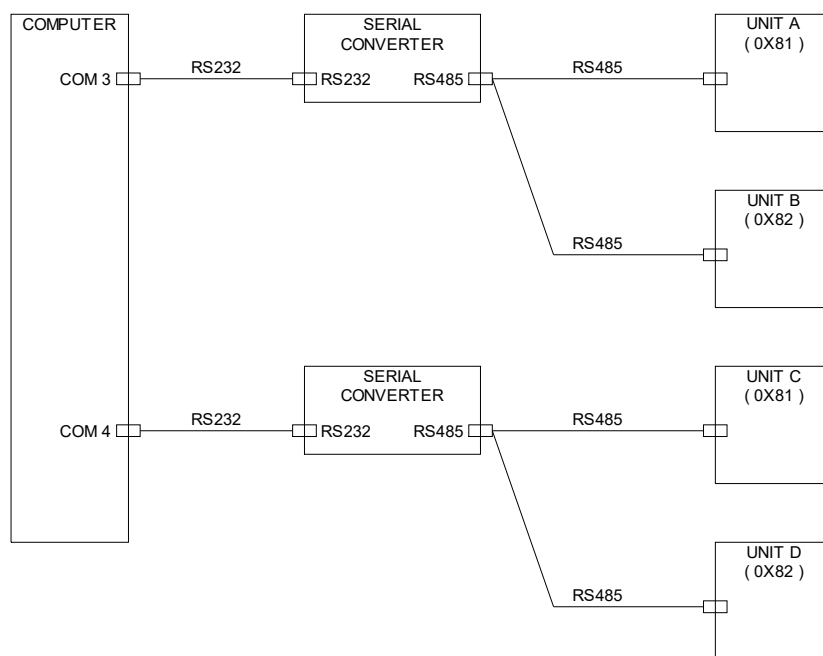


Figure 6. Multiple Units Connected to Multiple Ports

5. Maintenance

The AccuPositioner itself has no user-serviceable parts. It should not be opened or disassembled outside of the factory.

The mounting block and yoke bracket may be removed for cleaning or replacement. See Section 7, and the procedures below.

5.1. Remove and Replace the Yoke Bracket and Yoke Ring

Refer to Figure 9.

1. Remove and save the two screws (item 70) attaching the yoke bracket (item 20) to the yoke ring (item 30).
2. Remove and save the four screws (item 70) attaching the yoke bracket (item 20) to the positioner shaft (item 10).
3. Reverse the procedure to replace the yoke bracket.

5.2. Remove the Mounting Bracket and Stop Ring

NOTE

The mounting bracket and stop ring must be replaced in the correct orientation. If the orientation is wrong when the parts are reinstalled, the unit will not have the full travel.

Refer to Figure 9.

1. Drive the pan axis of the unit to the middle of its travel.
2. Place the stop collar, item 40, on the shaft with the disk pin opposite the housing pin (Figure 7).
3. Center the slot of the mounting bracket, (item 50) on the stop collar pin, and attach item 50 to the shaft using the 4 screws (item 70). See Figure 8.

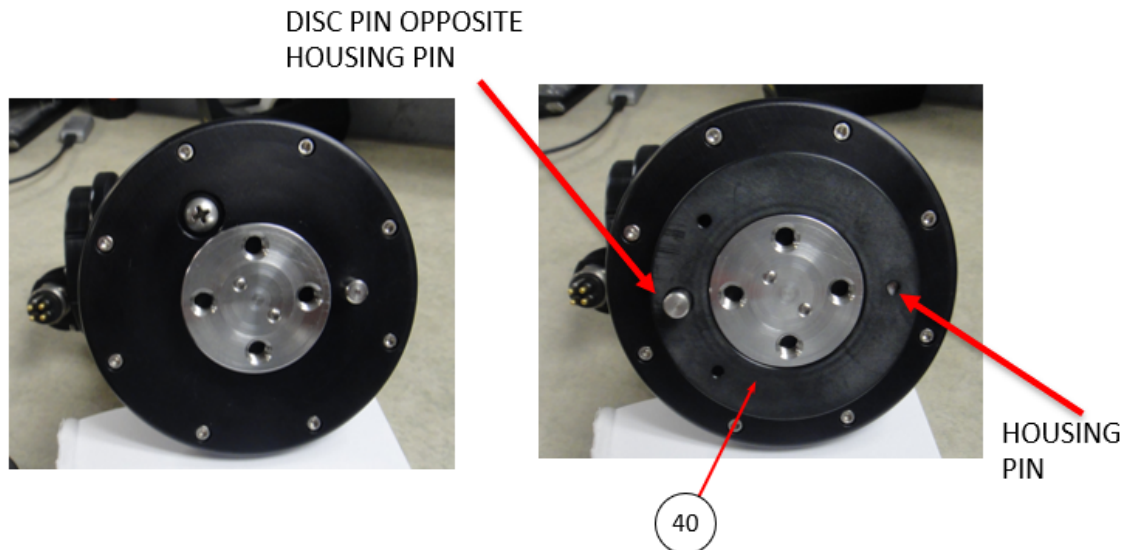
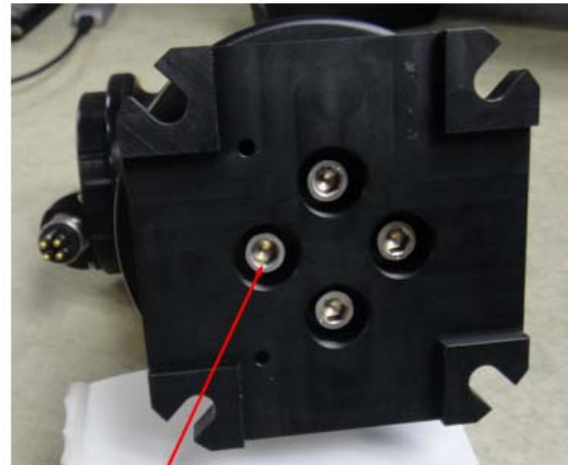


Figure 7. Stop Collar Install



50

MOUNTING BRACKET SLOT
CENTERED ON DISC PIN



70

Figure 8. Mounting Bracket Install

6. Troubleshooting

This section provides guidance for those problems with the AccuPositioner most likely to be encountered.

CAUTION

Troubleshooting should be performed only by trained technicians.

6.1. Systematic TroubleShooting Procedure

NOTE

The pinout in this manual is standard and is the pinout for most AccuPositioners. It may be necessary to verify the pinout of custom units.

1. Verify the unit is connected correctly based on the pinout in Figure 3.
2. Verify the power supply on the power pin pin 2 is 24V.
3. Verify the data pins are connected to the computer via an RS485 converter.
 - a. The RS485 side of the converter is toward the unit.
 - b. The RS232 side is toward the computer.
4. Verify the computer recognizes the RS485 converter
 - a. Open *Device Manager*
 - b. Expand (*Ports COM & LPT*)
 - c. Plug and unplug the converter from the computer to observe which COMPORT appears and disappears. This is the comport of the unit.
 - d. Leave the converter plugged in
5. Switch on power to the AccuPositioner. Verify the unit draws .5 - .7 amps and cycles brief short distances.
6. Use Helios (Helios units) or Eos (Visca Units) to try to find the unit.

NOTE

Any familiar serial terminal emulator program may be used for the following steps. The guidance here is for RealTerm.

7. Use RealTerm to try to find the unit (Helios)
 - a. Open RealTerm
 - b. Under the Display Tab/Display As, select *ASCII*
 - c. Under the Port Tab, Select the COM Port from step 4
 - d. Under the Send Tab, enter A?000
 - e. Click Send ASCII
 - f. The unit should respond with the unit info
8. Use RealTerm to try to find the unit (Visca)
 - a. Open RealTerm
 - b. Under the Display Tab, select Hex[Space]
 - c. Under the Port Tab, Select the COM Port from step 4
 - d. Under the Send Tab, enter A?000
 - e. Click Send ASCII
 - f. The unit should respond with the unit inf

6.2. Unit Slips or drops the load

If the unit slips at a lower torque than expected, consider the following guidance.

1. Check the maximum acceleration. Very high maximum acceleration has a small effect on the torque felt by the unit, and some effect on the torque capacity of the motor.
2. Check the maximum velocity. The unit should be able to provide 10 ft lbf at 24 deg/s. It may provide slightly more than this at lower speeds.
3. Check the load calculations. Aside from the weight, ensure that acceleration, and drag from water current, vehicle motion, and load rotation have been taken into account. These must be directly added to the torque from the mass of the load.
4. Note that the unit has zero holding capacity when it is not powered.

6.3. Unit Does Not Respond to Commands

If the unit does not respond to serial commands, consider the following guidance.

1. If the unit does not draw current, contact Remote Ocean Systems for guidance.
2. If the unit does not operate but draws approximately 0.5 A, use the Eos GUI to aid in the following steps
 - a. Verify the unit is installed and connected properly (Section 3.1), with the RS485 connection connected to the USB port of a computer with Eos available.
 - b. Turn the power to the unit off and on again.
 - c. Open the Eos GUI and check whether the unit is detected.
 - d. If the unit is detected, attempt to move the unit with the GUI. If the unit moves as-expected, the problem lies in the user software or firmware.
 - e. If the unit does not operate as expected through the GUI, open the log window. The log window will list attempts to communicate on all available com ports. Inspect the list for the com port / address that the unit should be on. The log should show whether the unit sent a partial response, no response at all, or an incorrect response. The log window may also indicate that the unit responded from a different address than expected.
 - f. The GUI also provides a diagnostic window which gives the ability to send formatted commands on a selected comport and address. Compare any responses received to the expected responses listed in Table 5.
 - g. If the unit still does not communicate, contact Remote Ocean Systems.

6.4. Unit Rotates in Unexpectedly Short Range of Travel

If the unit rotates normally, but for a shorter than expected range consider the following guidance

1. Verify the soft stops are set as expected.
2. Remove the mounting bracket and attempt to rotate the unit again. If the unit rotates normally without the mounting bracket, the hard stop is not installed correctly. See Section 5.1.

7. Parts List

Table 7 contains a list of parts available for use with the AccuPositioner. Figure 9 is an exploded view.

Table 7. Parts List

Item	Qty	Part No.	Description
20	1	21-30307-01	BRKT YOKE P15 AL
30	1	21-30330-01	BRKT RING YOKE W/INSERT P15 DL
40	1	21-30360-01	RING STOP MNTG P15 ACETAL
50	1	21-30361-01	BASE MNTG P15 DL
70	10	62-20133-4	SCREW SHHX 0.25-20X0.5 IN 316SS
80	1	62-20214-4	PIN DOWEL 0.25 IN DIA X 0.5 IN LG 316SS
90		60-02043-4	VIBRA-TITE THREAD LOCKING COMPOUND

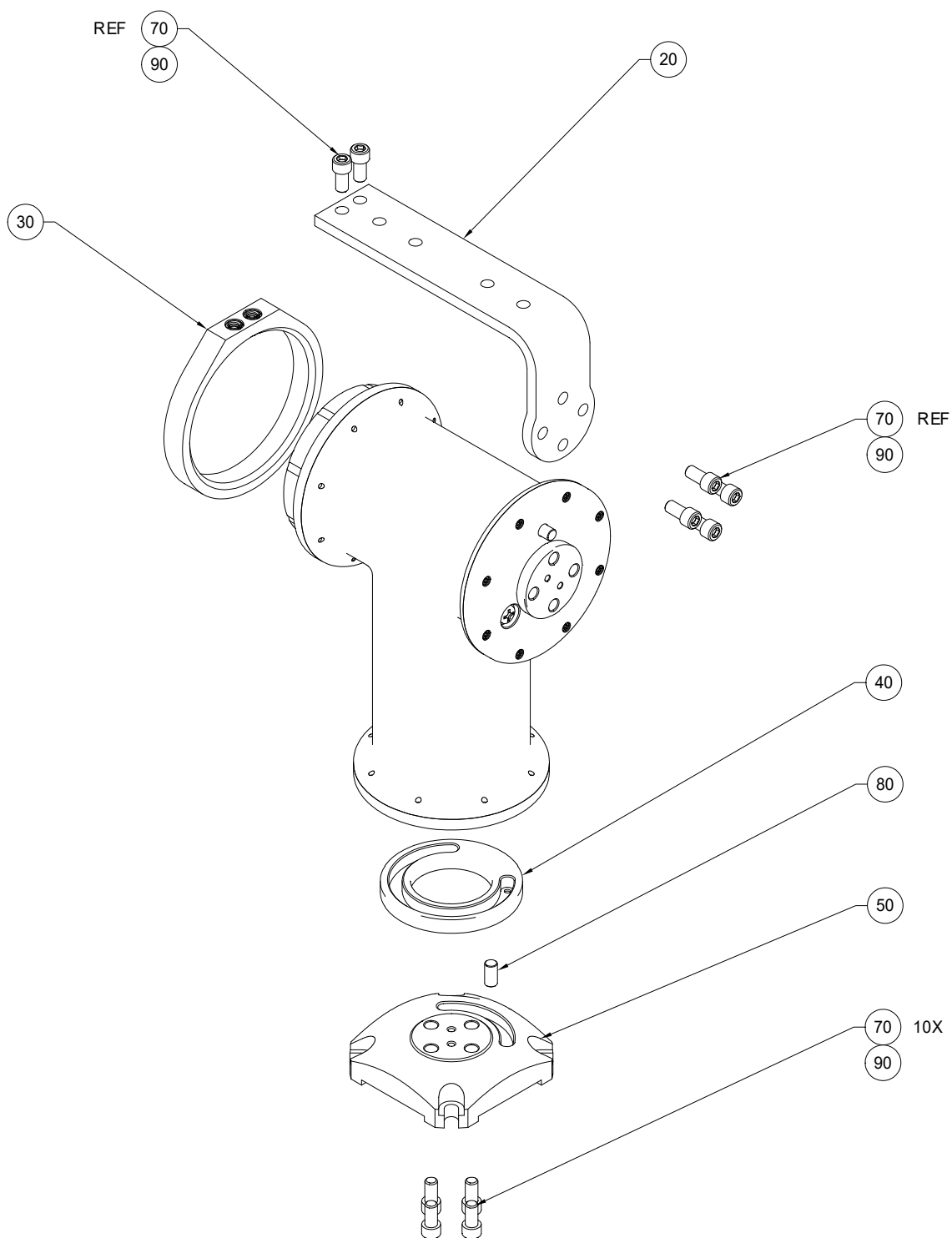


Figure 9. Exploded View